

AQA

(Predicted Paper by mrdoylemaths.co.uk)

Please write clearly in block capitals.

Surname _____

Forenames (S) _____

GCSE MATHEMATICS

Foundation Tier

EST. 2023
MR DOYLE MATHS
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Paper 3 Calculator

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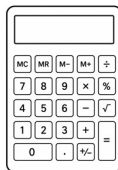
2026 Predicted Paper 3

Time allowed: 1 hour 30 minutes

Materials

For this paper you have:

- A calculator
- Mathematical instruments
- The Formula Sheet



Disclaimer

This resource has been designed using key topics based on questions that came up in AQA Paper 1 and 2 exams from this year. It is intended as a high-quality revision aid to support exam preparation.

However, it should not be treated as an exact prediction of any future exam paper. While some of the topics included may appear in the real exam, others may not. No revision resource including this one can guarantee what will or will not appear regardless of claims seen online or on social media.

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Answer **all** questions in the spaces provided.

1 Here is a list of numbers

5 9 8 20 3

1 (a) Which numbers are even?

[1 mark]

Answer _____

1 (b) Which number is a multiple of 20?

[1 mark]

Answer _____

1 (c) Which number is a prime number?

[1 mark]

Answer _____

1 (d) Which number is a square number?

[1 mark]

Answer _____

2 Write down all the factors of 30

[2 marks]

Answer _____

- 3 Five students pay a total of £45 for a meal.

How much did they pay on average?

[2 marks]

Answer £

- 4 Simplify fully $\frac{4x^6}{2x^4}$

[2 marks]

Answer

- 5 Factorise fully $15x - 12$

[1 mark]

Answer

- 6 A number y is rounded to 1 decimal place.

The result is 3.9

Write down the error interval for y

[2 marks]

Answer

 $\leq y <$

7 The total price of a taxi journey is calculated by

a fixed charge of £5
and
a charge of £2.50 per mile

7 (a) Write a formula to work out the total cost, $£C$, of a journey of m miles.

[2 marks]

$£C =$ _____

7 (b) Work out the total cost of a 4-mile journey.

[2 marks]

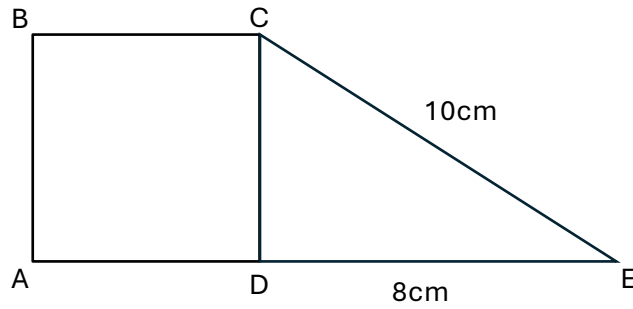
Answer £ _____

7 (c) If the total cost is £10, how many miles was the taxi ride?

[2 marks]

Answer _____ miles

- 8 ABCD is a square and CDE is a rectangle



Not drawn
accurately

Find the perimeter of the square ABCD.

[4 marks]

Answer _____ cm

- 9 Make m the subject $m^2 - 7 = n$

[2 marks]

$m =$ _____

☞

10 In 2000, the population of Nottingham was 745,000.

In 2020, the population of Nottingham was 834,400.

Work out the percentage change in population.

Give your answer to 1 decimal place.

[3 marks]

Answer _____ %

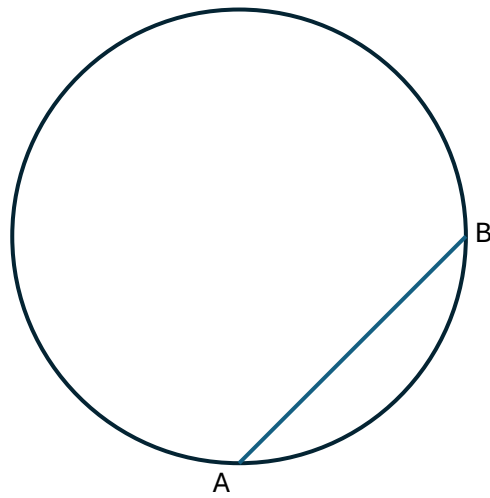
11 A van travels at 3 miles in 15 minutes.

Work out the average speed of the car in miles per hour.

[3 marks]

Answer _____ mph

- 12** Here is a circle with points A and B on the circumference.



Not drawn
accurately

- 12 (a)** What is the mathematical name of the line AB?

[1 marks]

Answer _____

- 12 (b)** The area of the circle is 100cm^2 .
Find the radius of the circle to 1 decimal place.

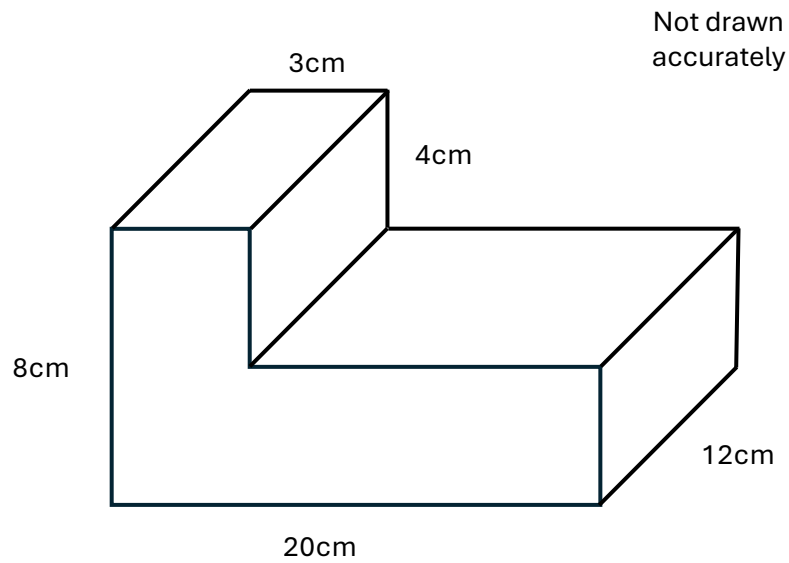
[3 marks]

Answer _____ cm

13

Find the volume of the prism.

[4 marks]



Answer _____ cm^3

- 14** 110 people visit a museum.
60 are adults, the rest are children.

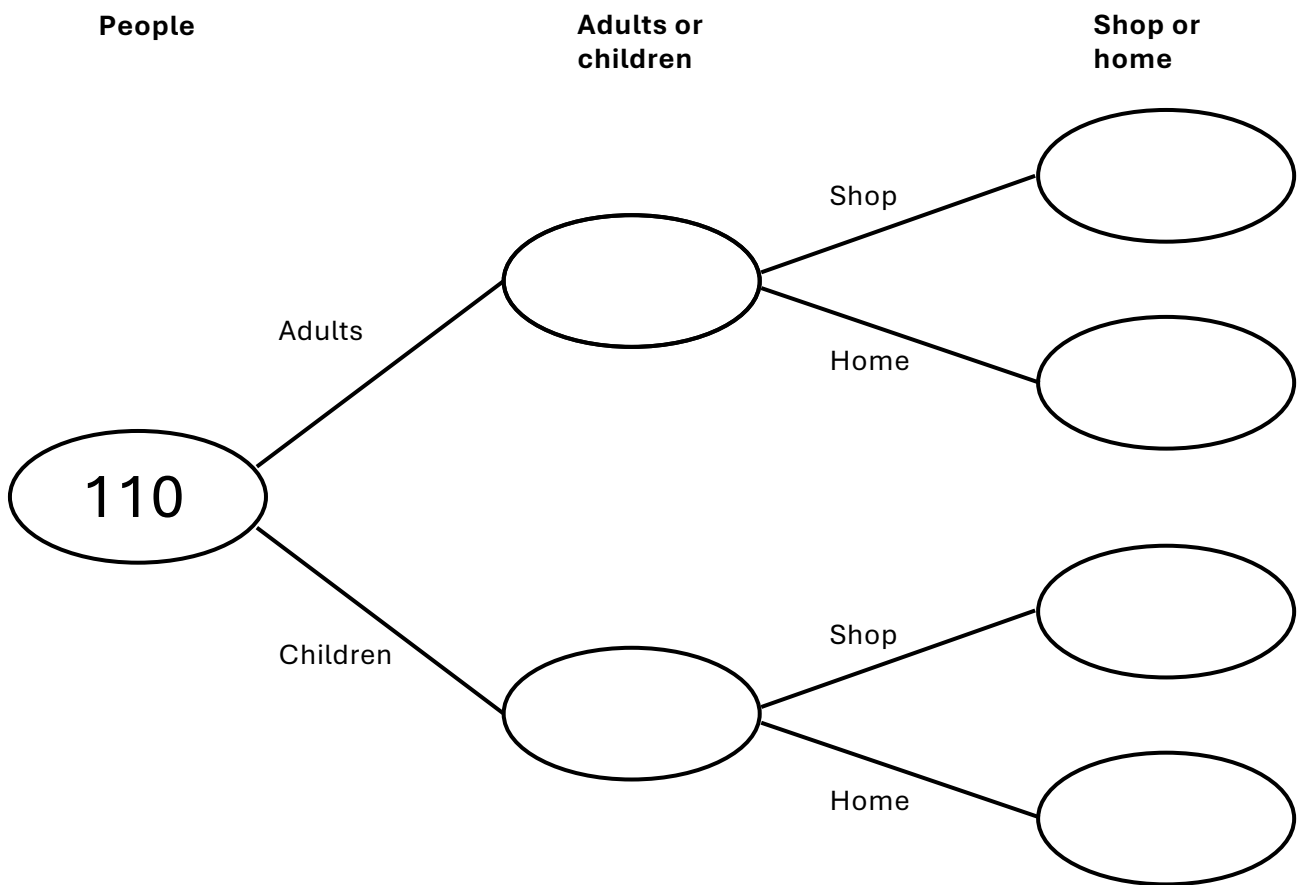
When visitors are leaving, they can visit the museum shop or go home.

40% of adults visit the shop.

In total 75 people visit the shop.

- 14 (a)** Complete the frequency tree.

[4 marks]



- 14 (b)** What fraction of adults visit don't visit the shop?

Give your answer in its simplest form.

[2 marks]

Answer

- 15** A Fibonacci-type sequence begins

4 -5

The sequence continues by adding the two previous terms

Work out the next **two** terms.

[2 marks]

Answer _____ and _____

- 16** The table shows information about the number of minutes 60 people spend watching TV in a typical day.

Time spent watching TV, T (minutes)	Frequency		
$0 < T \leq 20$	5		
$20 < T \leq 30$	10		
$30 < T \leq 40$	20		
$40 < T \leq 60$	25		

Work out an estimate for the mean time spent watching TV.

[4 marks]

Answer _____ minutes

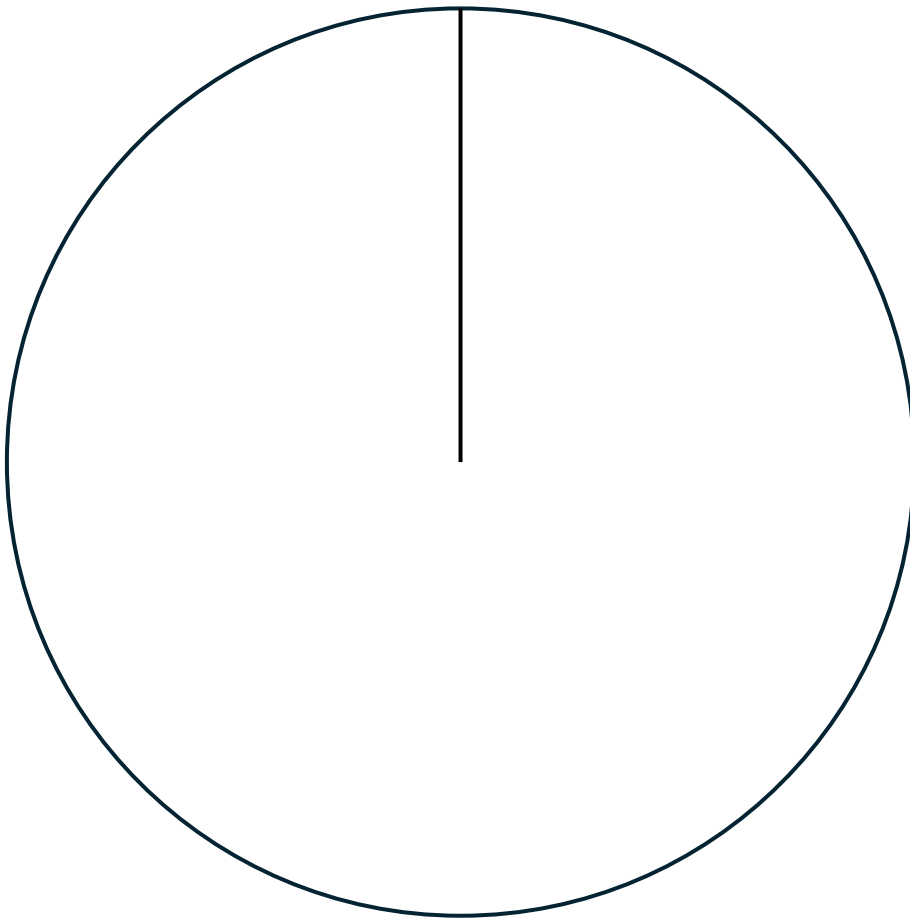
17 60 students in a school were asked their favorite lesson.

The table below shows the data.

Result	Frequency
Drama	15
English	10
History	5
Mathematics	30

Draw an accurate pie chart using the table.

[3 marks]



- 18** Here is a table of values for the equation $y = 4x - 2$

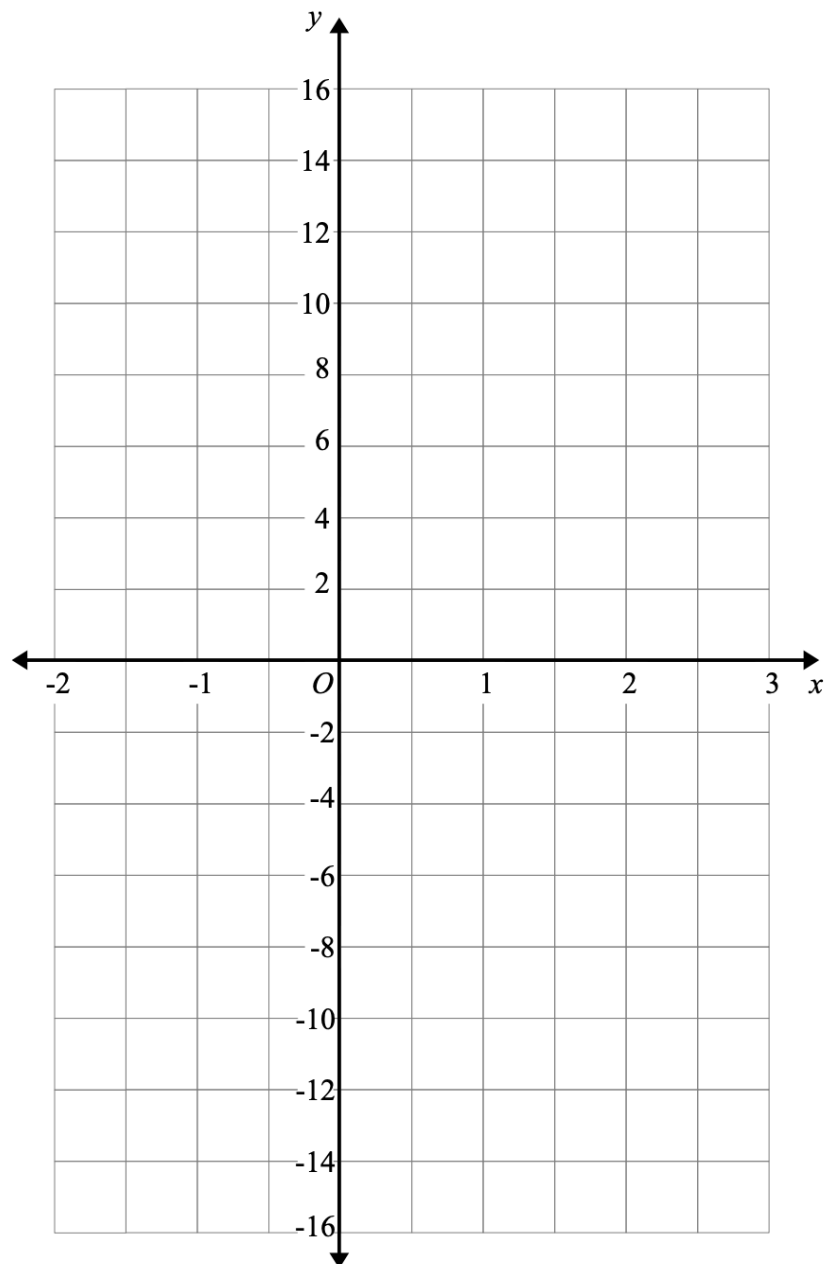
x	-2	-1	0	1	2	3
y				2		

- 18 (a)** Complete the table of values.

[2 marks]

- 18 (b)** Draw the graph of $y = 4x - 2$ for values of x from 1 to 4

[2 marks]



19 Michael drives from London to Manchester.

Michael leaves London at 8:15 am.

Michael drives for $3\frac{1}{4}$ hours before stopping for a break.

The break lasts for 30 minutes.

Michael then drives for another minutes to reach Manchester.

What time does Michael arrive in Manchester?

[3 marks]

Answer _____

20 A bag contains only blue and green discs.

number of blue discs : number of green discs = 7 : 2

20 (a) Write down the fraction of green counters in the bag.

[1 mark]

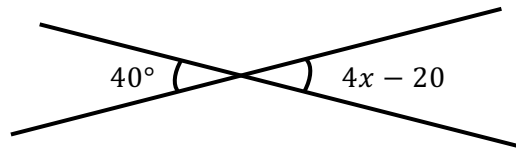
Answer _____

20 (b) Write the ratio of blue to green discs in the form 1 : n

[1 mark]

Answer _____

- 21 Here is a table of values for the equation $y = 4x - 2$



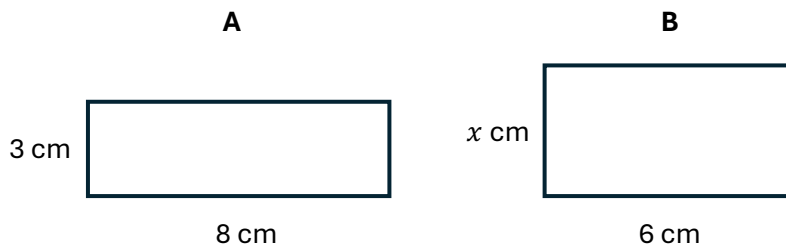
Not drawn
accurately

Work out the value of x

[2 marks]

Answer _____

- 22 The two rectangles have the **same area**.



Not drawn
accurately

Work out the ratio Perimeter A : Perimeter B

[4 marks]

Answer _____ : _____

23 Michael drives from London to Manchester.

Michael leaves London at 8:15 am.

Michael drives for $3\frac{1}{4}$ hours before stopping for a break.

The break lasts for 30 minutes.

Michael then drives for another minutes to reach Manchester.

What time does Michael arrive in Manchester?

[3 marks]

Answer _____

24 A bag contains only blue and green discs.

number of blue discs : number of green discs = 7 : 2

24 (a) Write down the fraction of green counters in the bag.

[1 mark]

Answer _____

24 (b) Write the ratio of blue to green discs in the form 1 : n

[1 mark]

Answer _____

25 The original value of a bike is £1240

The value of the bike decrease by

25% in the first year

then

10% in the second year

Work out the value of the bike after these two years

[3 marks]

Answer £ _____

26 80 students were asked how they get to university.

They either walk, cycle or travel by car.

$\frac{5}{8}$ of the students walk.

students who cycle : students who travel by car = 1 : 2

How many students travel by car?

[3 marks]

Answer £ _____

There are no questions printed on this page

*Do not write
outside the
box*